Adaptive Frequency-Domain Soft-Thresholding for Noise-Robust Image Classification

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***Abstract*—We propose a lightweight test-time preprocessing method based on adaptive frequency-domain soft-thresholding for noise-robust image classification. The method estimates corruption strength from high-frequency Fourier magnitudes using a robust median-based statistic and suppresses noise-sensitive spectral components while preserving phase. All experiments are conducted on CIFAR-10. The proposed nonlinear formulation improves robustness under multiple noise types while preserving clean accuracy with minimal latency overhead.**

I. Introduction

Deep neural networks are highly sensitive to input corruption. Even moderate noise can significantly degrade classification performance. While robustness can be improved through retraining, such approaches are costly and impractical when a pretrained model must remain fixed.

We instead focus on test-time preprocessing. Classical filters such as bilateral and median filtering provide partial robustness, but they are not uniformly effective across corruption types. In particular, median filtering is effective for impulsive corruption, whereas bilateral filtering is often more suitable for smoother additive corruption.

We propose a frequency-domain approach that suppresses noise-dominated spectral components while preserving structural information encoded in phase. All experiments are conducted on CIFAR-10, and all reported quantitative and qualitative results correspond strictly to this dataset rather than to ImageNet-scale evaluation.

Our main contributions are as follows:

- We propose an adaptive test-time preprocessing method based on frequency-domain soft-thresholding.
- We introduce a nonlinear threshold formulation that improves the clean–robustness trade-off.
- We provide evaluation on CIFAR-10 under Gaussian, speckle, salt-and-pepper, and Poisson noise.
- We analyze why the method works, how sensitive it is to threshold parameters, and how it compares in terms of efficiency.

II. Method

A. Formulation

Let

$$f_\theta : \mathbb{R}^{C\times H\times W} \to \mathbb{R}^K \tag{1}$$

be a pretrained classifier. Given a noisy input $x^{(n)}$, we apply a preprocessing operator $\mathcal{P}$ and classify the processed input:

$$\hat{y} = \arg\max_k f_\theta(\mathcal{P}(x^{(n)}))_k. \tag{2}$$

B. Fourier Representation

We transform the noisy image into the frequency domain:

$$X = \mathcal{F}(x^{(n)}). \tag{3}$$

Its magnitude and phase are given by

$$M = |X|, \qquad \Phi = \angle X. \tag{4}$$

C. Adaptive Thresholding

We estimate a robust corruption scale using the median absolute deviation (MAD):

$$m = \mathrm{median}(M), \tag{5}$$

$$\mathrm{MAD} = \mathrm{median}(|M - m|), \tag{6}$$

$$\hat{\sigma} = \frac{\mathrm{MAD}}{0.6745}. \tag{7}$$

The adaptive threshold is defined as

$$\tau = \alpha\hat{\sigma}^\gamma, \tag{8}$$

where α controls threshold strength and γ controls nonlinearity. We then apply soft-thresholding:

$$M' = \max(M - \tau, 0). \tag{9}$$

D. Reconstruction

The processed spectrum is reconstructed using the original phase:

$$X' = M' \odot e^{j\Phi}, \tag{10}$$

and the final image is obtained by inverse Fourier transform:

$$x' = \mathcal{F}^{-1}(X'). \tag{11}$$

E. Linear and Nonlinear Variants

When $\gamma = 1$, the threshold becomes linearly proportional to the estimated corruption level:

$$\tau_{\mathrm{lin}} = \alpha\hat{\sigma}. \tag{12}$$

When $\gamma > 1$, the threshold grows nonlinearly with the estimated corruption level, allowing stronger suppression under severe corruption while remaining less aggressive in cleaner cases.

III. Why It Works

The proposed method works because it aligns three useful properties of noisy image spectra: frequency selectivity, phase preservation, and adaptive shrinkage.

A. High-Frequency Magnitudes Are More Noise-Sensitive

Many common corruptions introduce disproportionately strong perturbations in high-frequency components. Gaussian and speckle noise spread energy broadly across the spectrum, while salt-and-pepper noise produces sparse but sharp discontinuities that also manifest strongly in high-frequency magnitude components. By contrast, semantically meaningful image content is not uniformly distributed across all frequencies.

This means that directly suppressing all frequencies would be undesirable, but selectively attenuating noise-dominated high-frequency energy can improve robustness without discarding the entire signal.

B. Phase Preservation Retains Structure

A key design choice of the proposed method is that it modifies only the magnitude spectrum while preserving phase. In natural images, phase carries much of the structural layout, including edge alignment, contour geometry, and coarse object arrangement. If both magnitude and phase were aggressively altered, denoising could easily destroy class-relevant structure. Preserving phase therefore acts as a structural anchor during preprocessing.

C. Soft-Thresholding Is More Stable Than Hard Removal

The proposed method uses soft-thresholding rather than hard-thresholding. Hard removal would set all coefficients below a threshold to zero and keep all others unchanged, creating abrupt spectral discontinuities. Soft-thresholding instead shrinks magnitudes continuously:

$$M' = \max(M - \tau, 0). \tag{13}$$

This produces smoother spectral modification and reduces the risk of introducing artifacts or excessive distortion in borderline coefficients.

D. Why Nonlinear Scaling Helps

The most important part of the design is the nonlinear threshold

$$\tau = \alpha \hat{\sigma}^{\gamma}. \tag{14}$$

If $\gamma = 1$, the threshold grows linearly with the estimated corruption scale. However, corruption severity does not necessarily translate into classifier fragility in a linear manner. Under stronger noise, a linear threshold may remain too weak, leaving substantial corruption intact. By setting $\gamma > 1$, the threshold grows super-linearly and becomes more aggressive only when the estimated corruption is large.

This gives the method two desirable properties:

- under mild corruption, thresholding remains conservative and does not overly damage clean structure;
- under stronger corruption, thresholding becomes sufficiently strong to suppress a larger fraction of noise-sensitive coefficients.

E. Why the Method Balances Clean and Noisy Performance

The overall success of the method comes from balancing robustness and preservation. Bilateral and median filtering are effective in particular regimes, but they use fixed spatial operations that can oversmooth images or specialize too strongly to a specific corruption type. Our method instead adapts its spectral suppression strength to the estimated corruption level of each input. This explains why it remains competitive across multiple noise regimes while keeping clean accuracy almost unchanged.

IV. Experimental Setup

All experiments are conducted on CIFAR-10. We use a pretrained ResNet-18 backbone and apply the proposed preprocessing only at test time. The classifier parameters remain fixed during evaluation.

We consider clean inputs and multiple corruption settings, including Gaussian, speckle, salt-and-pepper, and Poisson noise. This allows us to evaluate both clean preservation and robustness across diverse noise characteristics.

V. Evaluation Metrics

Let $\{(x_i, y_i)\}_{i=1}^{N}$ denote the evaluation set. Top-1 accuracy is defined as

$$\mathrm{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(\hat{y}_i = y_i) \times 100. \tag{15}$$

For a corruption operator $\mathcal{N}$, corrupted-input accuracy is

$$\mathrm{Acc}_{\mathcal{N}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left(\arg\max_k f_\theta(\mathcal{P}(\mathcal{N}(x_i)))_k = y_i \right) \times 100. \tag{16}$$

We also measure average inference latency

$$\mathrm{Lat} = \frac{1}{N} \sum_{i=1}^{N} t_i, \tag{17}$$

where t_i includes preprocessing and classifier forward propagation.

VI. Results

We first present the main comparison across representative noise conditions.

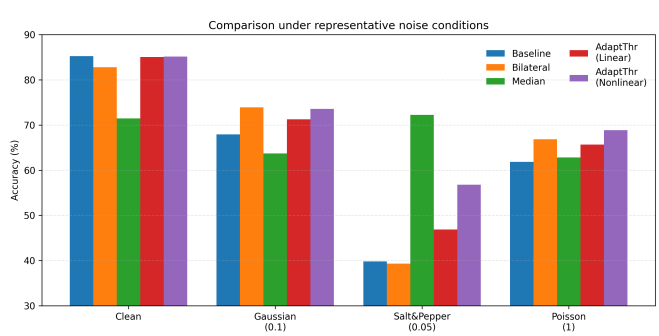

Fig. 1. Comparison across clean, Gaussian, salt-and-pepper, and Poisson noise on CIFAR-10.

Figure 1 shows that the nonlinear method preserves clean accuracy while improving robustness under Gaussian and Poisson noise. It also improves substantially under salt-and-pepper noise, although median filtering remains strongest in that specific impulsive-noise setting.

We next examine performance across a broader set of corruption types and severities.

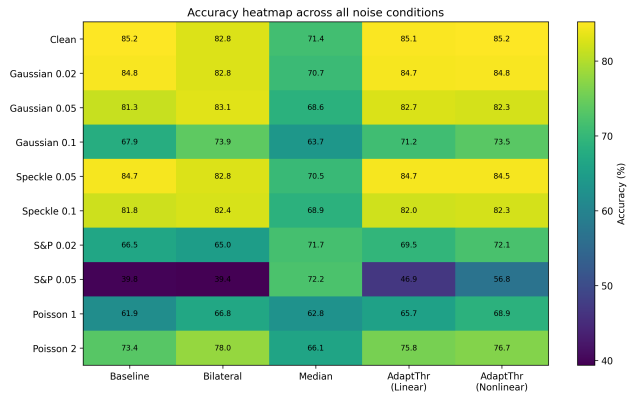

Fig. 2. Accuracy heatmap across multiple noise types and severities on CIFAR-10.

The heatmap in Fig. 2 shows that the proposed nonlinear method performs consistently across diverse conditions rather than specializing in a single case.

A central design goal of the paper is to improve noisy-input performance without sacrificing clean accuracy.

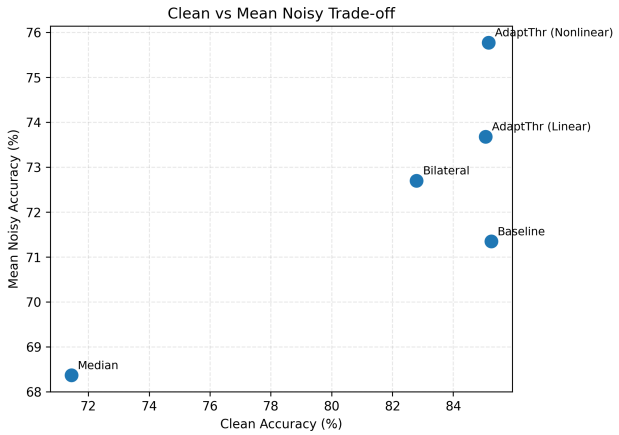

Fig. 3. Clean vs. mean noisy trade-off on CIFAR-10.

As shown in Fig. 3, the nonlinear method achieves the best trade-off between clean and noisy accuracy.

We further analyze computational efficiency.

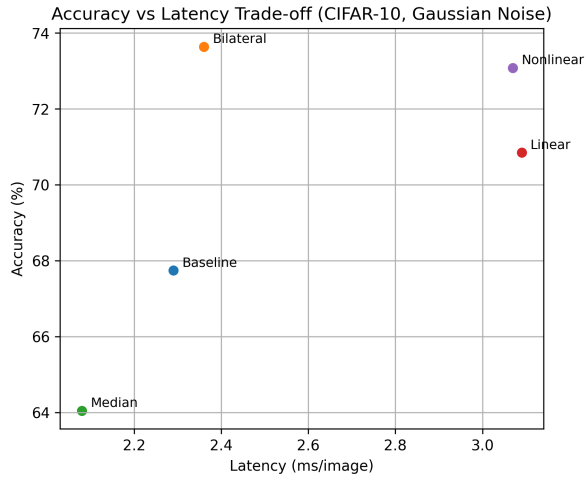

Fig. 4. Accuracy vs. latency trade-off on CIFAR-10 under Gaussian noise.

Figure 4 shows that the proposed method improves robustness with only a small computational overhead.

Finally, we isolate the effect of nonlinear thresholding.

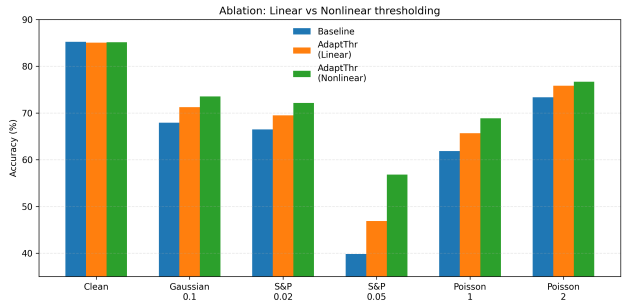

Fig. 5. Ablation comparing linear and nonlinear thresholding.

Figure 5 shows that the nonlinear formulation consistently outperforms the linear variant, particularly under stronger corruptions.

VII. Threshold Sensitivity Analysis

The proposed method relies on the adaptive threshold

$$\tau = \alpha \hat{\sigma}^{\gamma}. \tag{18}$$

To better understand the effect of threshold design, we performed a fine-grained search over α, γ, the high-frequency ratio h_f, and the channel-wise option.

The best nonlinear configuration was found at

$$\alpha = 1.0, \qquad \gamma = 1.8, \qquad h_f = 0.65, \qquad \texttt{channel_wise} = \texttt{False}.$$

TABLE I
Best nonlinear configuration from fine-grained search.

α	γ	h_f	Channel-wise	Mean Noisy Acc. (%)
1.0	1.8	0.65	False	75.66

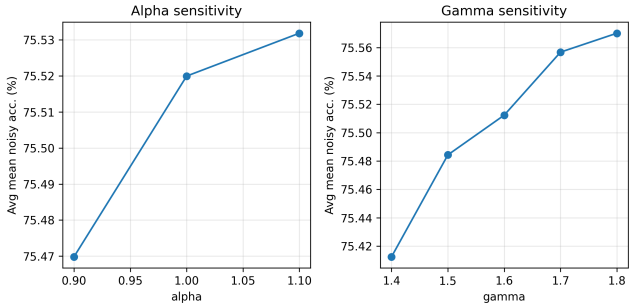

Fig. 6. Sensitivity of the proposed nonlinear threshold to α and γ. Each point is averaged over the remaining threshold-search settings.

Under the best configuration, the nonlinear method achieved 85.16% clean accuracy and 75.77% mean noisy accuracy. By comparison, the baseline achieved 85.25% clean accuracy and 71.35% mean noisy accuracy. Thus, the nonlinear formulation improves mean noisy accuracy by 4.42 percentage points while introducing only negligible clean-performance degradation.

Figure 6 shows the sensitivity of the method to the threshold parameters α and γ. For α, performance is relatively stable in the range from 1.0 to 1.1, indicating that the method is not overly sensitive to precise threshold scaling. For γ, performance improves as the exponent increases from smaller values and reaches its best region around 1.7–1.8. This trend supports the use of nonlinear scaling rather than a purely linear threshold.

We also compare the clean and mean noisy accuracy across all methods.

TABLE II
COMPARISON OF CLEAN AND MEAN NOISY ACCURACY.

Method	Clean Acc. (%)	Mean Noisy Acc. (%)
Baseline	85.25	71.35
Bilateral	82.79	72.70
Median	71.44	68.37
AdaptThr (Linear)	85.06	73.68
AdaptThr (Nonlinear)	85.16	75.77

The comparison between linear and nonlinear thresholding is particularly important. The linear variant achieves 73.68% mean noisy accuracy, whereas the nonlinear variant reaches 75.77%. Therefore, nonlinear scaling improves mean noisy accuracy by 2.09 percentage points relative to the linear rule while also slightly improving clean accuracy.

These results indicate that the exponent γ is not a minor implementation detail. Rather, it enables the threshold to adapt more effectively to strong corruption, which leads to a more favorable clean–robustness balance. In addition, several nearby parameter settings also achieve similar overall scores in the fine search, suggesting that the method is not overly sensitive to a single hand-picked configuration.

VIII. EFFICIENCY

We summarize per-image inference latency in Table III. The proposed nonlinear method is slightly slower than direct baseline inference, but the absolute overhead remains small.

TABLE III
LATENCY ON CIFAR-10 (MS/IMAGE).

Method	Latency
Baseline	2.29
Bilateral	2.36
Median	2.08
Linear	3.09
Nonlinear	3.07

More specifically,

$$\Delta\text{Lat} \approx 3.07 - 2.29 = 0.78 \text{ ms}. \tag{19}$$

This latency increase is modest relative to the robustness gains shown earlier.

IX. QUALITATIVE ANALYSIS

We next provide qualitative evidence under two complementary noise regimes: Gaussian noise and salt-and-pepper noise.

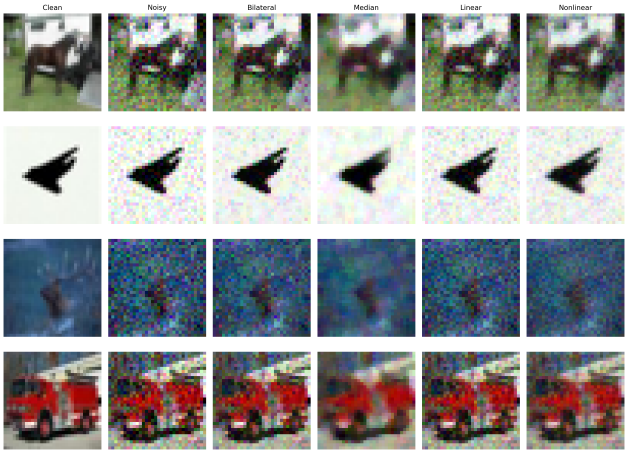

Fig. 7. Qualitative comparison under Gaussian noise.

Figure 7 shows that the nonlinear method reduces noise while preserving structure, avoiding the stronger blur introduced by some alternative filters.

Fig. 8. Qualitative comparison under salt-and-pepper noise.

Figure 8 shows that median filtering remains strongest for impulsive noise, but the proposed nonlinear method provides a better overall balance across different corruption types.

X. Discussion

The experimental results reveal several important patterns. First, classical filters remain strong in their preferred regimes: bilateral filtering is competitive under several Gaussian and Poisson settings, while median filtering dominates under strong salt-and-pepper noise. Second, these filters are less balanced overall, as reflected in their lower clean accuracy or lower mean noisy accuracy.

In contrast, the proposed nonlinear method provides the strongest overall clean–robustness trade-off. It does not dominate every single corruption setting, but it consistently remains competitive and achieves the highest mean noisy accuracy among the compared methods while almost preserving baseline clean performance. This makes it attractive as a general-purpose test-time robustness module rather than as a specialized denoiser for one particular corruption type.

XI. Conclusion

We presented a frequency-domain soft-thresholding method for noise-robust classification. The method improves robustness while preserving clean accuracy and introducing only minimal computational overhead in the CIFAR-10 setting. The threshold sensitivity study further shows that nonlinear scaling is a key ingredient for achieving a strong clean–robustness trade-off.

References

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